

CHEA ROTH

DATA SCIENCE AND MACHINE LEARNING

Engineering Degree in Apply Mathematics and Statistics

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SUMMARY

Innovative Data Scientist & ML Engineer with a strong academic foundation in Applied Mathematics and Statistics. Specialized in the end-to-end development of predictive systems, from feature engineering to scalable model deployment. Proven capability in architecting sophisticated ML frameworks and utilizing advanced visualization techniques to solve high-complexity technical challenges. Passionate about pushing the boundaries of AI research to optimize real-world performance.

TECHNICAL SKILLS

Programming & Tool

- Python, SQL, HTML, CSS, Matlab, React JS, R, Java, C#
- Git, Jupyter Notebook, Google Colab, VS Code, Anaconda
- Excel, Google Sheets, Data Studio, Docker,

Machine Learning & AI

- Regression, Classification, Random Forest, XGBoost, SVR
- LSTM, SARIMAX, Hybrid ML Models
- Scikit-learn, TensorFlow, Keras, LightGBM, CatBoost

Time Series & Forecasting

- ARIMA, SARIMA, LSTM, Prophet
- Exponential Smoothing, Statsmodels, PyFlux
- Trend & Seasonality Analysis

Data Analysis & Visualization

- EDA, Feature Engineering, Data Cleaning
- Matplotlib, Seaborn, Plotly, Tableau, Power BI
- Pandas, NumPy

Natural Language Processing (NLP)

- SentencePiece, Tokenization, Corpus Creation, LoRA Fine-Tuning
- Hugging Face Transformers, NLTK, SpaCy, Gensim

Database Management

- MySQL, PostgreSQL, SQLite, Microsoft Access, MongoDB (basic)
- SQL Queries, ERD Design, Relational Modeling

Other Strengths

- Statistics, Research Writing, Communication, Data Storytelling, Generative AI, Prompt Engineer
- Problem Solving, Critical Thinking, Business Insight

PROFESSIONAL EXPERIENCE

SchniecTech Group | *Battery Energy Storage System Engineer, Data Analysis*

Dec 2025 – Present

- Analyzed high-frequency time-series data (Active Power, Frequency, SOC, Voltage, Reactive Power) from EMS / SCADA systems
- Built and maintained interactive dashboards for real-time monitoring of grid and plant performance
- Correlated grid frequency with power dispatch behavior to detect instability patterns
- Monitored battery State of Charge (SOC) and optimized charge/discharge cycles
- Integrated NCC dispatch commands with plant response for operational alignment
- Conducted anomaly detection for power fluctuations, voltage deviations, and system faults
- Automated daily operational reports to support engineering and management teams

Ministry of Planning, Cambodia | *Data Scientist*

Feb 2025 – Oct 2025

- Developed a hybrid inflation forecasting model (XGBoost + LSTM + SARIMAX), improving accuracy of national economic projections.
- Built and maintained data pipelines for macroeconomic indicators, including cleaning, feature engineering, and stationarity testing.
- Conducted analysis of global (oil, gold) and domestic sector factors affecting inflation.
- Created interactive dashboards and reports to support policy-making and economic planning.
- Collaborated with economists and policymakers to translate complex data into actionable insights.

Jan 2024 – Feb 2025

- Taught EDA, statistics, machine learning, forecasting, and data visualization with a focus on practical Python and Power BI projects.
- Guided students in building dashboards, performing analysis, and deriving actionable insights from real datasets.
- Designed hands-on exercises and mini-projects to strengthen students’ analytical, problem-solving, and coding skills.
- Mentored students on data storytelling and presentation, preparing them to communicate insights effectively to technical

SELECTED PROJECT

Laptop Price Prediction Role: Data Analyst & Model Developer Tools: Python, Pandas, Scikit-learn, BeautifulSoup, Matplotlib, Seaborn • Performed data cleaning, exploratory data analysis (EDA), and feature engineering. • Applied multiple regression models including Linear Regression, Random Forest, and Support Vector Regressor (SVR). • Achieved high prediction accuracy with the Support Vector Regressor model.	Hotel Reservation System Database Role: Database Designer & Developer Tools & Skills: MySQL, SQL, ERD Design, Relational Database Modeling • Designed and implemented a database system to manage hotel operations such as room booking, client management, and staff scheduling. • Built ER diagrams and relational schemas for key entities Client, Staff, Room, Service, Floor, Booking. • Applied data normalization, primary and foreign key constraints to ensure data integrity. • Supported receptionist workflows including booking verification and payment confirmation.
Forecasting Tourism in Cambodia Role: Data Analyst / Time Series Forecaster Tools & Skills: Python (Pandas, NumPy, Matplotlib), ARIMA, SARIMA, Prophet, LSTM, Time Series Modeling • Collected and preprocessed tourism data for time series analysis. • Evaluated model performance using Mean Squared Error (MSE) and residual diagnostics. • Identified seasonal patterns and recovery trends post-COVID-19. • Recommended data-driven insights for tourism recovery planning and policy-making.	Crop Yield Prediction & Recommendation Role: Data Analyst Tools & Skills: Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Jupyter Notebook • Collected and cleaned agricultural datasets. • Built machine learning models (Random Forest, XGBoost) to predict crop yield. • Engineered key features from soil and weather data. • Evaluated model performance using R², MAE, and RMSE. • Developed a recommendation system for suitable crop selection.

EDUCATION

Institute of Technology of Cambodia (ITC) 2020 – 2025 Engineering Degree in Data Science Major in Applied Mathematics and Statistics Thesis: Analysis and Forecasting of Inflation in Cambodia	Aii Language Center 2019 – 2022 English French (Basic)
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REFERENCE

PHAUK SOKKHEY, Ph.D Position: Deputy head of department of applied mathematics and statistics (AMS) (អនុប្រធានដេប៉ាតឺម៉ង់គណិតវិទ្យាអនុវត្ត និងស្ថិតិ) and Head of Master Program of Data Science Phone: 012 939 310 Email: phauk.sokkhey@itc.edu.kh	Mr. Pakrina Long Position: IT Lectureer in DATA Science and Institute of Technology Cambodia Phone: 010 64 44 96 Email: long.pakrigna@itc.edu.kh
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